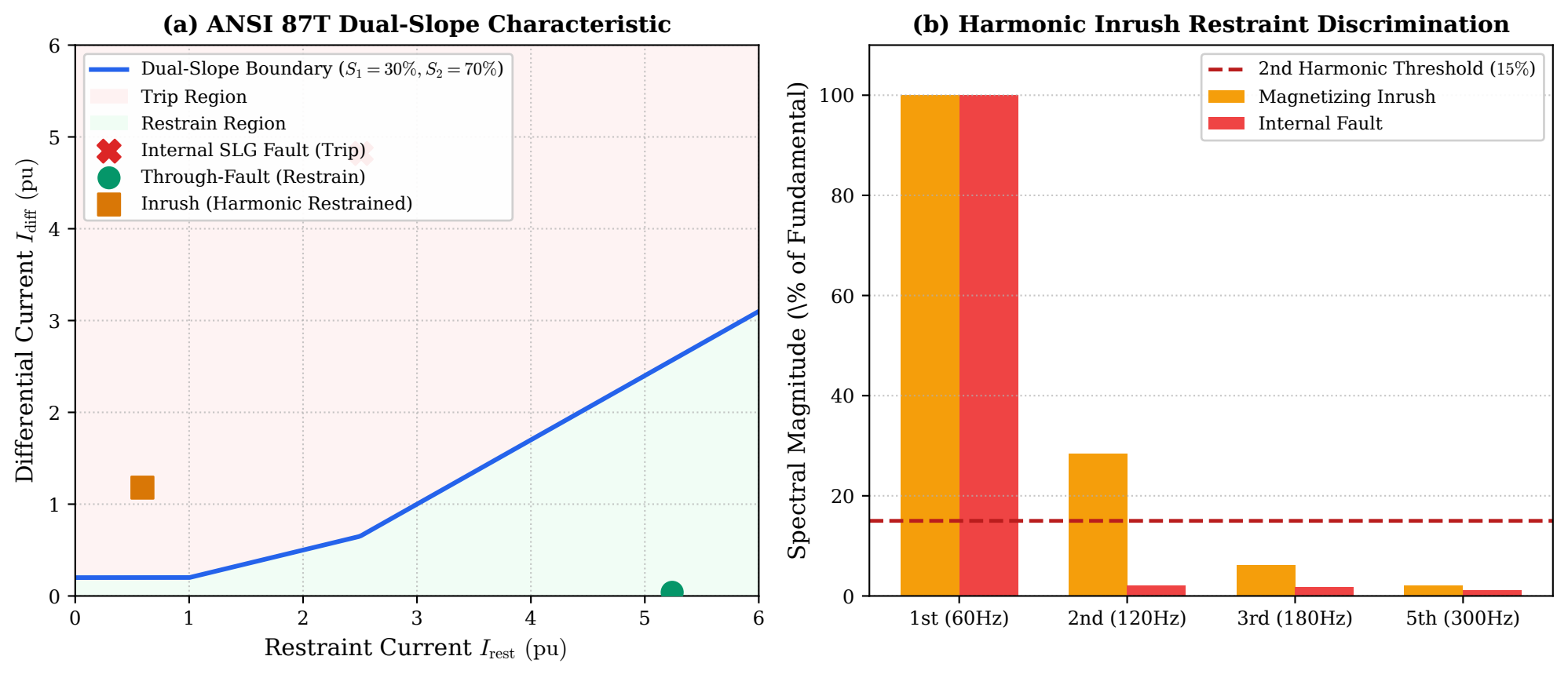

(a) ANSI 87T Dual-Slope Characteristic
Dual-Slope Boundary ($S_1 = 30\%$, $S_2 = 70\%$)
Trip Region
Restrain Region
Internal SLG Fault (Trip)
Through-Fault (Restrain)
Inrush (Harmonic Restrained)
Differential Current I_{diff} (pu)
Restraint Current I_{rest} (pu)
0
1
2
3
4
5
6
(b) Harmonic Inrush Restraint Discrimination
2nd Harmonic Threshold (15%)
Magnetizing Inrush
Internal Fault
Spectral Magnitude (% of Fundamental)
0
20
40
60
80
100
1st (60Hz)
2nd (120Hz)
3rd (180Hz)
5th (300Hz)